

ComputerSeekho – Student Admission & Placement Management System

Overview

ComputerSeekho is a full-stack web application developed for **SMVITA** to digitalize and automate the complete student admission and placement lifecycle. The system replaces manual paperwork with an efficient online platform, allowing staff to manage student inquiries, admissions, payments, batches, and placements from a single dashboard.

The project follows a structured workflow where every student progresses through different stages—from submitting an inquiry to finally getting placed in a company. It also provides a public website for prospective students and a secure admin portal for institute staff.

Project Workflow

The system manages the entire student journey through the following pipeline:

```
graph TD; Inquiry --> FollowUp[Follow-Up]; FollowUp --> StudentRegistration[Student Registration]; StudentRegistration --> CourseEnrollment[Course Enrollment]; CourseEnrollment --> FeePayment[Fee Payment]; FeePayment --> BatchManagement[Batch Management]; BatchManagement --> CourseCompletion[Course Completion]; CourseCompletion --> Placement;
```

Every registration is linked to an inquiry, ensuring complete traceability throughout the student's lifecycle.

Public Website Features

The public portal provides all information required by prospective students and includes:

- Home page with banners and announcements
 - Course information
 - Student inquiry form with validations
 - Automatic confirmation email after inquiry submission
 - News & Events
 - Testimonials
 - Image Gallery
 - Recruiter showcase
 - Placement statistics
 - Batch albums
 - Contact information
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Admin Portal Features

The administration panel allows staff to efficiently manage every aspect of the institute.

Inquiry Management

- Create and manage student inquiries
 - Automatic counselor assignment
 - Inquiry validation
 - Follow-up scheduling
 - Inquiry status tracking
 - Close inquiries with predefined closure reasons
 - Hidden closed inquiries from active follow-up list
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Follow-Up Management

- Automatic first follow-up after inquiry creation
 - Record communication history
 - Update inquiry status
 - Schedule future follow-ups
 - Convert inquiry into registration
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Student Management

- Student registration
- Student profile management

- Address management
 - Student search
 - Student history
 - Linked inquiry records
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Course Management

- Course Categories
 - Course Details
 - Course Fees
 - Course Duration
 - Faculty Assignment
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Payment Management

- Automatic fee calculation
 - Two-installment payment structure
 - Payment history
 - Receipt generation (PDF)
 - Installment tracking
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Batch Management

- Create batches
 - Assign students
 - Faculty allocation
 - Batch schedules
 - Batch albums
 - Cover image management
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Placement Management

- Recruiter Management
 - Placement Drives
 - Student Placements
 - Recruiter-wise placed students
 - Placement Reports
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Master Data Management

Manage all lookup data including:

- Courses
- Course Categories
- Recruiters
- Staff
- Announcements
- Banners
- News & Events
- Gallery
- Testimonials
- Placement Drives
- Closure Reasons

Key Features

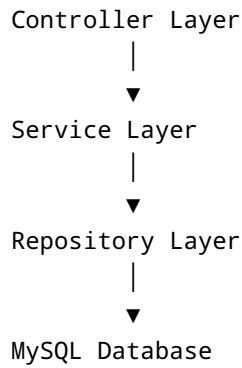
- Complete Admission Management
- Inquiry Tracking
- Automated Follow-Ups
- Student Registration
- Course Enrollment
- Fee Management
- PDF Receipt Generation
- Batch Management
- Placement Tracking
- Recruiter Management
- Dashboard Analytics
- Role-Based Administration
- Email Notifications
- File Upload Support
- Search & Filtering
- Data Validation

System Architecture

The application follows a layered architecture to ensure maintainability and scalability.

Presentation Layer





This separation of concerns keeps the code modular, reusable, and easy to maintain.



Core Modules

- Inquiry Module
- Follow-Up Module
- Student Module
- Course Module
- Enrollment Module
- Payment Module
- Batch Module
- Faculty Module
- Placement Module
- Recruiter Module
- Gallery Module
- Announcement Module
- Authentication Module



Business Rules

The application enforces several important business rules:

- Every student registration must originate from an inquiry.
 - Counselors are automatically assigned to new inquiries.
 - Closed inquiries require a mandatory closure reason.
 - Course fees are automatically populated during registration.
 - Payment is divided into two installments.
 - Each batch maintains its own image album.
 - Recruiters display all students placed in their company.
 - Closed inquiries are excluded from active follow-up lists.
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Tech Stack

Backend

- Java
- Spring Boot
- Spring Data JPA
- Spring Security
- Hibernate
- REST APIs

Database

- MySQL

Frontend

- React.js
- HTML5
- CSS3
- JavaScript
- Bootstrap

Tools

- Maven
- Git
- GitHub
- Postman
- IntelliJ IDEA / Eclipse



Benefits

- Eliminates manual paperwork
 - Improves admission workflow
 - Centralized student records
 - Faster follow-up process
 - Accurate fee tracking
 - Better placement management
 - Improved reporting and analytics
 - Enhanced data consistency through enforced business rules
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Future Enhancements

- Student Portal
 - Faculty Portal
 - Online Fee Payment Gateway
 - SMS Notifications
 - Mobile Application
 - Attendance Management
 - Certificate Generation
 - Advanced Analytics Dashboard
 - AI-based Student Insights
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Conclusion

ComputerSeekho is a comprehensive institute management system designed to automate the complete admission and placement process. By integrating inquiry management, student registration, payment tracking, batch administration, and placement management into a single platform, it significantly improves operational efficiency, ensures data integrity, and provides a seamless experience for both administrators and students.